Class 35: Semiconductors

In the previous class we have looked at the free electron approximation and the tight binding approximation. We found that both these approaches are able to explain anisotropy in material properties. They are therefore quite complete as models for materials.

In this class we will look at semiconductors and we will also see the utility of $E\ Vs\ k$ diagrams – especially the ability of the $E\ Vs\ k$ diagrams to explain material phenomena better than flat band diagrams.

We will begin by looking at semiconductors in general.

Semiconductors are a class of materials that display significantly less conductivity compared to metals. However, their conductivity can be controlled and manipulated in many technically interesting ways and hence useful devices can be made using these materials. They typically have a filled valence band, and an empty conduction band, with a band gap of less than 2 eV.

In semiconductors, with the availability of adequate amounts of energy, electrons will transition from the filled valence band to the empty conduction band. When such a transition occurs, the electron leaves behind an empty state in the valence band. Such empty states, which originally contained electrons, are called 'holes' and these are also treated as charge carriers with a positive charge, numerically equivalent to that of electronic charge. Conceptually, the situation in semiconductors can be thought of as follows: For current to exist, electrons must move, and for electrons to move they must find empty sites adjacent to them to which they can then move. When the valence band is full, the electrons in the valence band do not have vacant sites within the valence band to move to. At the same time, the conduction band is empty and there are no electrons available in the conduction band to support conduction as well. Therefore no electronic conductivity is possible. If sufficient energy, in the form of thermal energy or light for example, is provided, the energy enables a few electrons to transition from the valence band to the conduction band. The conduction band now contains a few electrons and a large number of vacant sites. The electrons are able to move in this situation, and are able to support current. Similarly, the large number of electrons that remain in the valence band also have a few vacant sites, or holes, that have been left behind by the electrons that transitioned to the conduction band. Therefore the electrons in the valence band can also move in response to an electric field, and support current. It turns out that it is easier to follow the conduction process in the valence band by focusing on the movement of the small number of holes, rather than the large number of electrons. Electrons moving in one direction is the same as the holes moving in the opposite direction. This approach of focusing on electrons in the conduction band, and holes in the valence band, is therefore commonly used in the discussion of semiconductors. This same approach will be used in the discussion here as well.

Semiconductors are broadly of two types, intrinsic semiconductors, and extrinsic semiconductors.

Intrinsic semiconductors: These are materials which have do not have any deliberately added dopants, and are made as pure as possible. Their conductivity is a function of temperature only.

They could be elemental or compound semiconductors. Elemental semiconductors are Group IVA elements of the periodic table such as Si, and Ge which have band gaps of approximately 1.1 eV and 0.7 eV respectively. Compound semiconductors consist of pairs of elements on either side of Group IVA. Elements of Group IIIA and Group VA, such as GaAs, and InSb can be combined in a 1:1 ratio to synthesize compound intrinsic semiconductors. These are referred to as III-V compounds. Similarly we can combine elements which are even further apart in the periodic table, on either side of Group IVA, such as Group IIB and Group VIA and synthesize compound intrinsic semiconductors. These are called II-VI compounds and CdS is an example of such a compound. However, it is not possible to go to Groups further and further away from Group IVA, and expect combination of elements to result in semiconductors. As the elements get more widely separated in the periodic table, the bonds between the elements tend to become more ionic and less covalent. In ionic bonds the electrons are significantly localized to specific atoms, and they do not display semiconducting behavior. Incidentally, it is worth noting that semiconductors can be discussed and described using bonds in the material as well as the band structure of the material. The description chosen is a matter of convenience, and essentially identical discussion will result in both cases.

Extrinsic semiconductors: These are primarily Group IVA elements with tiny quantities of other elements deliberately added as dopants. Based on the dopant chosen, there is typically either a slight excess of electrons available as charge carriers, in which case the material is called an 'n-type semiconductor', or there is a slight excess of holes available as charge carriers, in which case the material is called a 'p-type semiconductor'. Doping a IVA element with a VA element results in an n-type semiconductor, while doping the IVA element with a IIIA element, results in a p-type semiconductor.

The Ohm's law states:

$$J = \sigma\varepsilon$$

Since the flux of charge equals the product of the concentration of charge carriers, times the charge of the charge carrier, times the drift velocity corresponding to the charge carrier, we have:

$$J = \sigma\varepsilon = n|e|v_d$$

Where v_d is the drift velocity of the electrons.

Rearranging, we have:

$$\sigma = n|e|\left(\frac{v_d}{\varepsilon}\right)$$

Since $\left(\frac{v_d}{\varepsilon}\right) = \mu_e$, or the mobility of the electrons, we have:

$$\sigma = n|e|\mu_e$$

In an intrinsic semiconductor, for each electron in the conduction band, there is a hole in the valence band. Due to the vacant sites available, both of the bands are able to support the

conduction process. Electrons moving in one direction, is the same as holes moving in the opposite direction. Therefore:

$$\sigma = n|e|\mu_e + p|e|\mu_h$$

Where p is the concentration of holes in the valence band, and μ_h is the mobility of the holes.

In intrinsic semiconductors, $n = p$ since each electron in the conduction band has left behind a hole in the valence band. This is therefore called the intrinsic carrier concentration, n_i.

Therefore, for intrinsic semiconductors:

$$n = p = n_i$$

And,

$$\sigma = n_i|e|(\mu_e + \mu_h)$$

In general, $\mu_e > \mu_h$. This can be understood as follows: From the perspective of electrons, the electrons in the conduction band have a large number of vacant sites to move to, and are therefore able to move more easily in this 'open space'. Therefore electrons in the conduction band are more mobile, resulting in a larger μ_e. Relatively speaking, the electrons in the valence band are situated in an extremely 'crowded space' with a large number of other electrons, and very limited vacant sites, or holes, to move into. Therefore the electrons in the valence band are less mobile – and since we use holes to describe the charge transport in the valence band, this results in a smaller μ_h.

In the case of extrinsic semiconductors, at room temperature, the charge carriers are dominated by the contributions of the dopants in the system. Therefore for an n-type semiconductor,

$$n \gg p$$

Which results in

$$\sigma \approx n|e|\mu_e$$

Similarly, for a p-type semiconductor, at room temperature,

$$\sigma \approx p|e|\mu_h$$

The reason the dopants dominate the extrinsic semiconductor behavior is that doping shifts the Fermi energy of the material to a value corresponding to the dopant level. Electron transfer to or from this level and the adjacent band becomes very easy – or requires a lot less energy, compared to the energy required to transfer electrons from the valence band to the conduction band. Intrinsic behavior is not precluded in an extrinsic semiconductor, it is just that it is negligible at room temperature and has an impact only at relatively high temperatures. The Fermi energies of intrinsic semiconductors, n-type semiconductors and p-type semiconductors are shown in Figure 35.1 below.

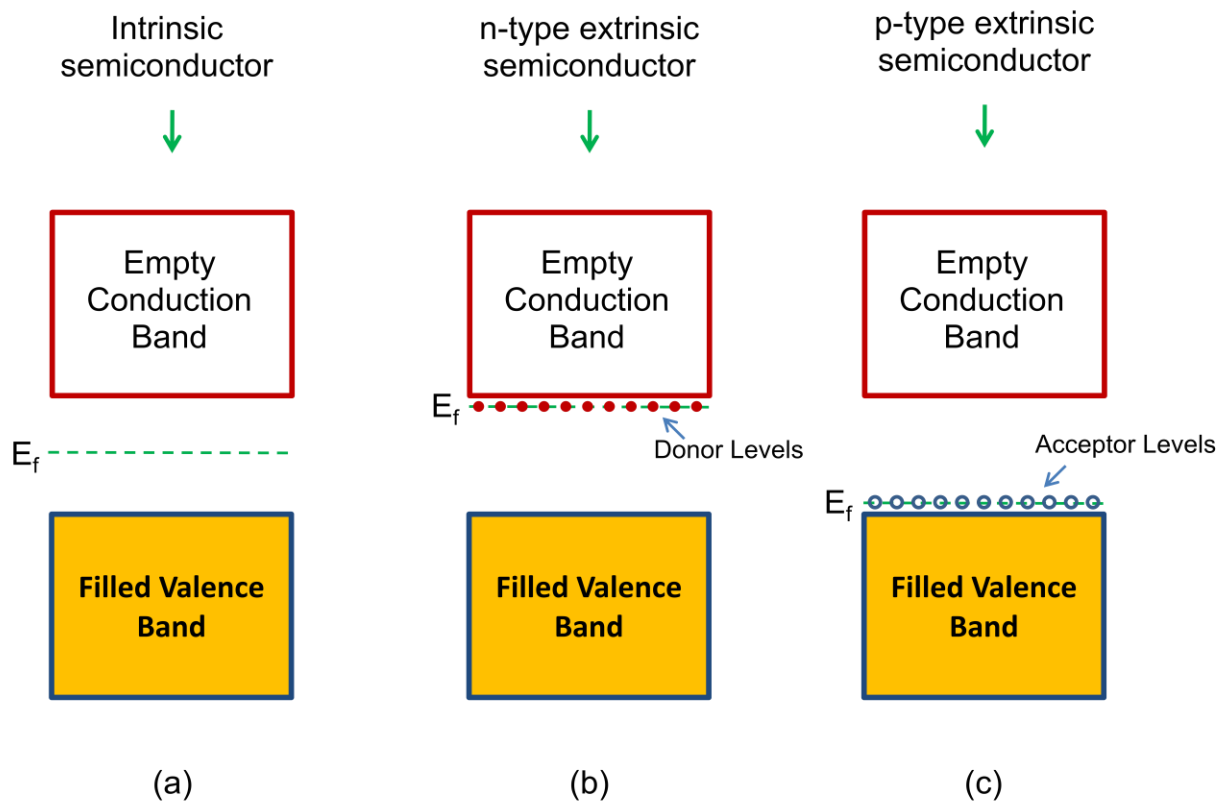


Figure 35.1: Fermi energy of (a) intrinsic, (b) n-type, and (c) p-type semiconductors.

The position of the Fermi energy in semiconductors is an important piece of information because when two semiconductors are brought in physical contact with each other, the value of the Fermi energy of the two materials decides the direction in which the electrons will flow to try and equalize the Fermi energies.

As noted earlier, conductivity depends on charge carrier concentration, the value of the charge, and the mobility of the charge. This is true regardless of whether the conductor is metallic or semiconducting. However, we see that metals show a positive thermal coefficient for resistivity, i.e. resistance increases with temperature, whereas for semiconductors the resistance decreases with temperature. Why does this occur?

The answer lies in the fact that for metals, the charge carrier concentration is largely constant as the temperature is increased. But the mobility of the electrons decreases on increasing temperature due to increased probability of collisions, due to the higher velocities of the electrons.

Therefore in the equation:

$$\sigma = N_f |e| \mu_e$$

As the temperature increases, the only parameter affected is μ_e, which decreases with increase in temperature. Therefore the conductivity decreases (or resistivity increases) with increase in temperature.

For semiconductors both the charge carrier concentration, as well as mobility of the charge carriers, are impacted by changes in temperature. The charge carrier concentration increases with temperature, and more than compensates for the decrease in the charge carrier mobility. Hence for semiconductors the conductivity increases with temperature, or the resistivity decreases with temperature.

The behavior of charge carrier concentration as a function of temperature, for semiconductors, is summarized in Figure 35.2 below.

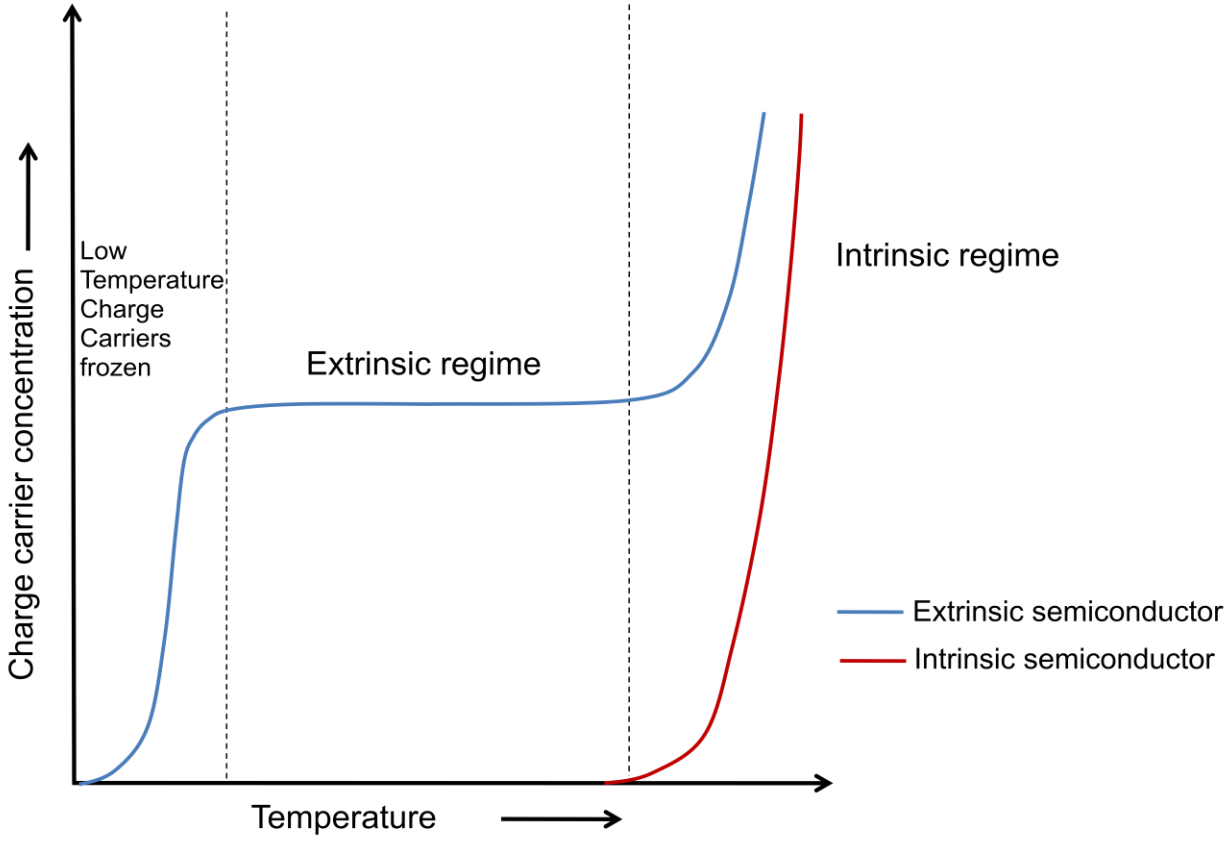


Figure 35.2: Charge carrier concentration for semiconductors, as a function of temperature.

For all semiconductors, at very low temperatures the charge carriers do not have enough energy to occupy vacant sites and hence the charge carrier concentration is very small. For extrinsic semiconductors, as the temperature increases, the dopant levels contribute to the charge carrier concentration even at relatively low temperatures, since the energy required to transfer the electrons to the conduction band from the donor level (for example), is very small. This contribution to the charge carrier concentration is fixed over a wide range of temperatures since the dopant concentration is fixed and the energy required is small. Only at much higher

temperatures, the intrinsic contributions begin to take effect with transitions from the host material's valence band to the conduction band across the band gap, which requires a relatively large amount of energy. For intrinsic semiconductors, this last contribution is the only one that is present. The intrinsic contribution increases with further increase in temperature since more electrons can make the transition at increased temperatures.

Having examined the basics of semiconductor behavior, we can now try to understand the optical properties of materials.

In general, when a material has a band gap of E_g, it can only absorb radiation that has enough energy to enable transition across the band gap. Therefore the minimum frequency, ν_{min}, of radiation that it can absorb is such that

$$E_g = h\nu_{min}$$

The material will not absorb frequencies lower than this, and will therefore be transparent to such radiations. This is also the reason that metals are opaque. In metals there is a continuous range of energy levels almost immediately above the Fermi energy. Therefore, even for very small frequencies, there are energy levels that can accept electrons gaining that small quantity of energy.

From the above discussion, we understand how materials absorb radiation. However, even after exceeding the threshold ν_{min}, it turns out that some materials with relatively low band gaps are quite ineffective in absorbing radiation, while other materials absorb more of the radiation incident on them. Why does this happen? Some process in the material impacts the probability of the absorption process and this information is not captured by the ν_{min} information alone.

The information that is not captured by just stating the band gap, is that in some cases the highest occupied energy level and the next lowest unoccupied energy level are not in the same location in k space. This implies that the electron not only has to gain energy greater than that corresponding to the band gap, but it also has to move to a different location in k space, to complete the transition. For an electron to gain energy when light falls on the material, it needs to interact with a photon of the appropriate energy. Since the photons are randomly striking the material, there is a probability associated with a given electron encountering a photon and gaining energy from it. If the next lowest unoccupied energy level is in the same location in k space, then interaction with the photon is all that is required for the electron to absorb the radiation and go to conduction band. Materials which have the highest occupied energy level and the next lowest unoccupied energy level in the same location in k space, are called '**direct band gap**' semiconductors. On the other hand if the next lowest unoccupied energy level is at another location in k space, the electron that has gained energy from the photon, must also encounter an appropriate lattice vibration to transport it to that location in k space. This encounter with an appropriate lattice vibration (lattice vibrations are referred to as phonons, and are discussed in greater detail in a later class) also has a probability associated with it, which is relatively low. The overall probability of such a material absorbing radiation is the product of the probability of the electron interacting with a photon of sufficiently high energy and the probability of the electron then encountering an appropriate phonon. This product of probabilities results in a

relatively low overall probability for the process. Therefore such materials are ineffective in absorbing radiation. Materials which have the highest occupied energy level and the next lowest unoccupied energy level at different locations in k space, are called '**indirect band gap**' semiconductors.

Direct band gap materials are therefore preferred for optical and opto-electronic applications, over indirect band gap materials.

The flat band diagrams do not distinguish between direct band gap materials and indirect band gap materials, while the $E\ Vs\ k$ diagrams do indicate the difference. Therefore, the $E\ Vs\ k$ diagram is much more capable in explaining material properties relative to flat band diagrams.

The difference between direct band gap and indirect band gap semiconductors, is highlighted in Figure 35.3 below.

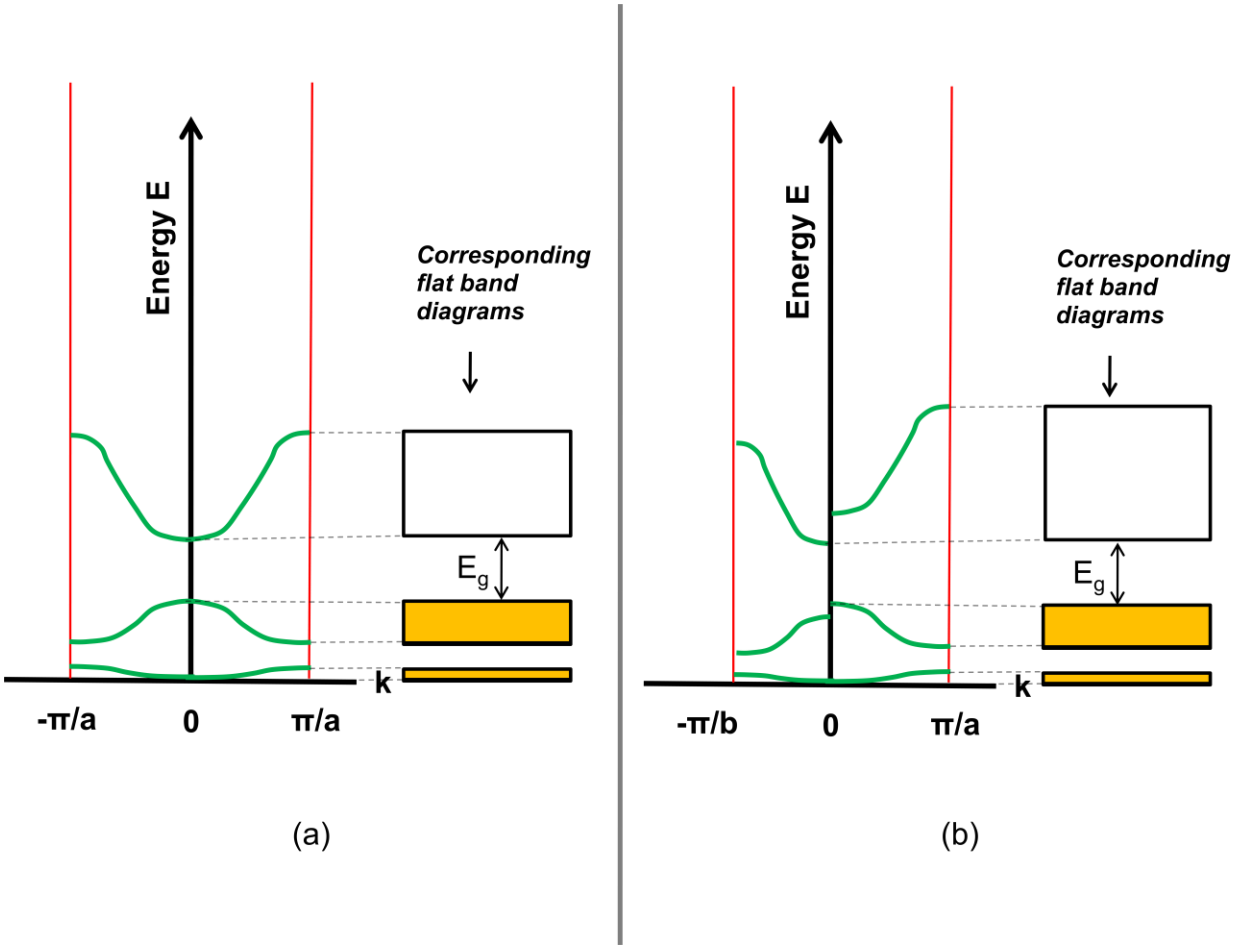


Figure 35.3: (a) Direct band gap semiconductor and (b) Indirect band gap semiconductor

In summary, in this class we have looked at semiconductors and at optical properties, and understood the utility of $E\ Vs\ k$ diagrams over and above that of flat band diagrams.